

Echo State Networks: Novel reservoir selection and hyperparameter optimization model for time series forecasting

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ABSTRACT

The use of computational intelligence models for multi-step time series forecasting tasks has presented satisfactory results in such a way that they are considered models with an excellent future for this type of problem. From the point of view of computational cost, the current alternatives combined with classical models are generating hybrid models that present even better results. Within the AutoML category, the optimization of hyperparameters and the selection of network topologies has become a challenge. Reservoir Computing, which is within the area of Recurrent Neural Networks (RNN), proposes a particular model called Echo State Networks, which has been tested in different applications with excellent results; however, the difficulty in specifying the hyperparameters has been the subject of continuous study given the random nature of the set of neurons called Reservoir. Based on the Separation Ratio Graph (SRG) model for performance analysis, this paper proposes a new model, called Echo State Network - Genetic Algorithm - Separation Ratio Graph (ESN-GA-SRG), which optimizes network hyperparameters and at the same time selects the best topology for the Reservoir using the SRG coefficient, to find the reservoir that offers the most suitable dynamic behavior. The performance of this new model is evaluated on forecasting two sets of time series benchmarks with different characteristics of sampling periodicity, skewness, and stationarity. The results obtained show that the ESN-GA-SRG model was superior in predicting these time series in most cases, with statistical significance, when compared to other models that have been presented for this type of problem in the literature.

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1. Introduction

The use of Recurrent Neural Networks (RNNs) models has expanded to several areas and applications, such as: pattern recognition [1]; prediction in mMIMO for enhanced performance in wireless communications [2]; trajectory tracking on soft continuum robots [3]; control for quadrotor attitude and altitude [4]; Data Analytics for Short Term Price and Load Forecasting in Smart Grids [5];, fault accommodation control for a biped robot [6];, discretizing time series [7]; table field extraction in business documents [8]; Lung cancer detection [9]; and recognition of English handwritten text [10], among others.

Through time, different types of Recurrent Neural Networks have been presented; the Hopfield Networks [11] were the starting point for the subsequent presentation of the first models of recur-

ring networks, such as the Jordan Networks [12], the bidirectional associative memory (BAM) [13], and the Elman Networks [14].

The Temporal Recurrent Neural Networks were presented by Dominey [15] as another model of recurring networks, demonstrating the potential of this type of network in solving several problems. However, the high computational cost required to carry out the training made them out of use by most researchers. The term “Reservoir Computing” (RC) was introduced to the scientific community as a new category of recurring networks that aims to explore the properties of these networks with a lower computational cost. Among the models in this new category, the Echo State Networks (ESNs) proposed by Jaeger [16], the Liquid State Machines (LSM) networks proposed by Natschläger and Maass [17] and the Decorrelation-Backpropagation Learning networks introduced by [18] are the most used networks in RC. Their main contributions are the reduction of the computational cost, with the use of simple training models. It is important to note that the Extreme Learning networks, although not considered part of the RC family, have an operating principle like the ESNs.

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It is noteworthy that the Echo State Networks (ESNs) were the networks that achieved the best performance in multi-step forecasting in the “Artificial Neural Network & Computational Intelligence Forecasting Competition - NN3”, transforming them into a potential model for several other applications, as can be seen in the publications below:

1. Control [19,20] and [21].
2. Time Series Forecast [22,23,24,25] and [26].
3. Pattern Recognition [27] and [30].
4. Systems Identification [29] and [30].
5. Modeling [31,32,33,34] and [35].
6. Classification, sorting, and ranking [36] and [37].

Thus, the main contribution of this article is to propose a new model, called Echo State Network - Genetic Algorithm - Separation Ratio Graph (ESN-GA-SRG), to automatically define the most suitable hyperparameters and reservoir of the network for the multi-step time-series forecasting. This model addresses the characteristics of AutoML model seeking to generate explainable and reproducible results.

The proposed methodology, supported by Genetic Algorithms, searches for the best hyperparameters configuration (size of the reservoir, connection percentage, among others). Therefore, through the controlled evaluation of the performance of the generated reservoirs, it is possible to choose a favorable configuration for the problem being tackled. The use of GA for the optimization of global hyperparameters in the configuration of the ESN network has been used in previous studies [39] and [40] with quite satisfactory results. In such cases, the parameters were different from those presented in this article. In addition, most of those previous works are based on randomly created connections in the reservoir, giving very different results, in addition to the difficulty in determining global parameters [41]. This paper proposes a method to select the most suitable reservoir to the problem at hand, extension of the SRG proposition for classification [42] for regression and time series forecasting.

This paper is divided into four additional sections. Section 2 briefly discusses reservoir computing models, considering training and evaluation metrics. Section 3 describes in detail the proposed ESN-GA-SRG model and its main characteristics. Section 4 presents the case studies for forecasting the NN3 and M3 series. Finally, the last section presents the conclusions and perspectives for new works.

2. Reservoir computing

RC is a type of recurrent neural network that, unlike Multi-Layer Perceptron (MLP) networks, show in the hidden layer a large set of neurons partially interconnected with each other and with connections to the input and output layer. They can also show connections from the output to the hidden layer (reservoir) as well as between the input layer to the output layer [43]. The fundamental principles of RC type networks can be summarized as:

- It uses a large number of neurons (known as reservoir), activated by means of input signals, and each unit of the reservoir creates its own nonlinear transformation of the input considering the transformation generated by the previous pattern. This process takes place in an environment with a much higher dimension than the one originally used [44].
- Output signals are obtained from an excited reservoir using, generally, a simple linear combination.

Temporal Recurrent Neural Networks (TRNNs) were introduced by Peter F. Dominey [15]. It was the pioneering model of Reservoir Computing (RC), although the term RC was not used at the time.

However, the methodology used for the design and training of TRNNs shares the same RC principles. From the moment that other models were developed, and the RC category was used for recurring networks with the principles described above, the TRNNs model was placed into this category. Presented as a case study, it used as a benchmark, the application “Complex Sensory-Motor Sequence Learning”. This model consists of two subnets: the first “prefrontal cortex (PFC)” is responsible for generating unique activity patterns that represent continuous states of the execution of a sequence; the second subnet, “Striatum”, associates the coded state patterns with a response for each execution point of the sequence.

Liquid State Machines were presented by Maass [45] as the results of research on recurring networks. As the operating principle, an analogy is made to the behavior generated by a pebble (inlet) when is thrown into a pond (liquid medium). In this analogy, ripples occur in the liquid medium, which are observed as a reaction (liquid state). The ripples are propagated by the surface of the water and have the possibility to interact with others that already exist. Therefore, the water would allow to maintain and integrate the information of the events that happened. When these ripples are captured by a camera (reading unit), they can provide information about the events, and thus associate these events with the observed ripples. Using a Liquid State Machine (LSM) network it is possible to transform a small temporal input into a larger dimension generated by liquid states. In addition, information about past stimuli is added to the current input signal.

Presented by Steil [18], the Decorrelation-Backpropagation (BPMC) training rule is applied to networks similar to ESNs or LSMs, with the difference that the reservoir units are 100% connected to each other. That is, all units are connected to all other units within the Reservoir. However, as with ESNs, only the weights of the output layer are adjusted. From the principle of delay, which must exist between the generated states and the inputs, it becomes possible for the output layer to combine these states to obtain the desired output.

2.1. Echo State networks

The Echo State Networks (ESNs) were presented by Herbert Jaeger [16], and became widely used for simplicity in training and the excellent results obtained in benchmarks such as “Japanese Vowel Competition” [46]) with an error rate of 0% in test set, and the “International Forecasting Competition NN3” [47], where it reached the second position, among 45 models, in the multi-step forecast of time series for the reduced version. This type of network was inspired by neuropsychological experiments [38], where one of the characteristics is the presence of at least one recurring connection, that is, the value obtained in the output layer at time n is used in the input layer and / or in the reservoir at time $n + 1$. Fig. 1 shows the connection scheme for ESN networks.

The most important element in ESNs is the reservoir, which contains many randomly interconnected and self-connected indoor units. The weights of the synapses of the reservoir neurons, after being initialized, remain unchanged. Only the weights of the output layer connections are modified during the training process. The weight update can be performed offline, using linear regression, or online using the recursive least squares method [49], offering low computational cost when trained offline.

2.2. Echo State networks training

The great advantage of ESNs over other recurrent networks in the RC area is the low computational cost when training is conducted offline. In this section, the status update and training pro-

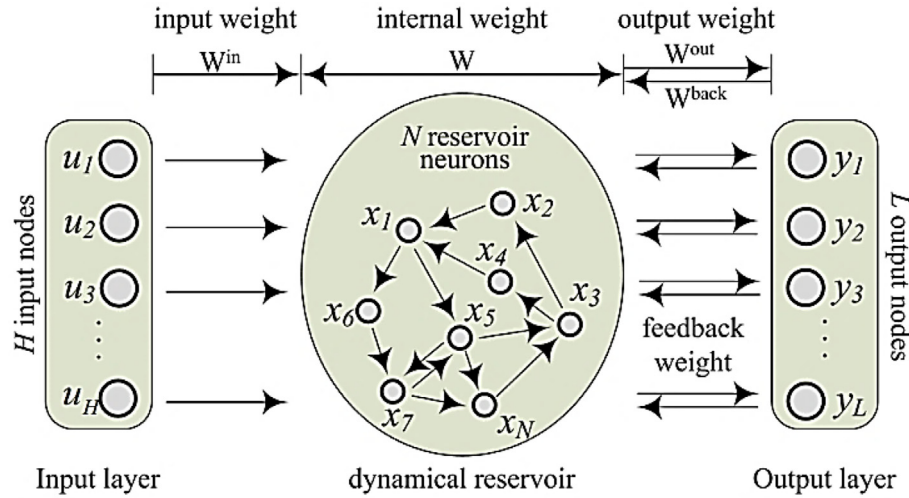


Fig. 1. RC. Adapted from [48]

cesses are presented in greater detail, as well as the main configuration characteristics.

The activation status of neurons in the reservoir is updated after each input pattern presented. Equation (1) presents all the elements that influence the activation status of reservoir neurons.

$$X(n) = (1 - \alpha)X(n-1) + \alpha\varphi(W^e U(n) + W^R X(n-1) + W^T D(n-1)) \quad (1)$$

where $X(n)$ is the internal state of the reservoir at time n ; α is the Leaking Rate, whose objective is to control the speed with which the states are updated when neurons of the Integrate-and-Fire type [50,51] are used; $\varphi()$ is the activation function of reservoir neurons; $D(n-1)$ is the potential generated in the output units in time $(n-1)$; $U(n)$ is the input vector; W^e , W^R and W^T are the weights for the input connections, Reservoir and recurrent connections of the output layer, respectively, which are randomly initialized with a uniform distribution.

In addition, he states that “the correct connection between the spectral ray and the Echo State property is that the Echo State property is violated if $\rho(W) < 1$ only in reservoirs that use the tanh activation function and that have no entry. In other configurations, the Echo State property can be obtained even when $\rho(W) > 1$ in systems with non-zero input, and it can also be lost even if $\rho(W) < 1$, although it is very difficult to build systems where this happens” [49]. [52] The outputs of the ESNs can be obtained using equation (2).

$$D(n) = \varphi^S(W^S) \cdot \begin{bmatrix} X(n) \\ D(n-1) \end{bmatrix} \quad (2)$$

where, φ^S can be a linear function or a sigmoid, and W^S is the matrix that contains the weights of the output connections, which are determined after training. There are models that can have connections from the input directly to the output, but the model used in this work uses the most usual configuration, that is, these connections between the input and the output are not considered.

Using K input units, N units in the reservoir and L units in the output layer, the dimensions of the weight matrices are: $W^e \in \mathbb{R}^{N \times K}$, $W^R \in \mathbb{R}^{N \times N}$, $W^T \in \mathbb{R}^{N \times L}$ and $W^S \in \mathbb{R}^{L \times N}$.

Offline training is carried out in two stages. First all patterns are presented only once and the Reservoir states are collected in the interval $[A_1, \dots, A_e]$, where A_1 e A_e are the start and end points of the sampling range (i.e. A_1 is the first one and A_e is the last standard

of the training set), respectively. These states are stored in a $[q, N]$ matrix J , as shown in equation (3).

$$J = \begin{bmatrix} x_{1,1} & \dots & x_{1,N} \\ \vdots & \ddots & \vdots \\ x_{q,1} & \dots & x_{q,N} \end{bmatrix} \quad (3)$$

In the training process, only the weights of the output layer W^S are adapted. Thus, after reservoir states have been collected, all desired outputs (targets) are collected in a $\tilde{D}$ matrix. Once matrices J and $\tilde{D}$ obtained are obtained, the weights of the output layer are acquired with the pseudo-inverse matrix method, according to equation (4).

$$W^S = (J^T J)^{-1} J^T \tilde{D} \quad (4)$$

The evaluation metric is usually the Mean Squared Error (MSE), presented in equation (5):

$$MSE = \frac{1}{(A_1 - A_e) + 1} \sum_{n=1}^e (y(n) - d(n))^2 \quad (5)$$

where $y(n)$ is the expected output and $d(n)$ is the actual output.

2.3. Reservoir status assessment metrics

Several techniques have been used to evaluate the quality of the states generated after the presentation of the standards for the networks that make up the CR area. In this section are presented three metrics considered as the most important according to [42], the last one being the result of the merger of the first two.

2.3.1. Separation coefficient

Initially, this metric was used for classification performing tasks networks. The separation coefficient is the average distance between the states generated by the reservoir after the presentation of the input vectors of each of the objective classes. This definition is so since the input vectors and the corresponding states are divided into discrete classes. Since all states of the reservoir, for inputs of the same class, must have the same output, it is expected that the distance between these states must be small; correspondingly, reservoir states of different classes must have a large distance. If this principle is fulfilled, the network will be able to perform the classification correctly. So, having a set of states x , all of the same class, $x_1, x_2, x_3, \dots, x_k \in O_j$, the mean of

the center of mass of the class O_j , $u(O_j(t))$, at time t , is defined by equation (6).

$$u(O_j(t)) = \frac{\sum_{i=1}^K x_i(t)}{K} \quad (6)$$

where, $x_i(t)$ represents the state at time t , K is the total number of states of class j and $u(O_j(t))$ is the center of mass of class O_j . Reservoir separation is specified in terms of two factors: the inter-class distance $C_d(t)$ defined in equation (7), and the intra-class distance $C_v(t)$ defined in equation (8).

$$C_d(t) = \sum_{m=1}^H \sum_{n=1}^H \frac{\|u(O_m(t)) - u(O_n(t))\|}{H^2} \quad (7)$$

$$C_v(t) = \frac{1}{H} \sum_{m=1}^H \frac{\sum_{i=1}^K \|u(O_m(t)) - x_i(t)\|}{k} \quad (8)$$

where, $u(O_m(t))$ and $u(O_n(t))$ are the centers of mass of the classes O_m and O_n and H is the number of classes. The separation coefficient is finally obtained using equation (9).

$$SEP_x(t) = \frac{C_d(t)}{C_v(t) + 1} \quad (9)$$

As presented in [53], the higher the separation coefficient, the greater the accuracy in classification, because the interclass distance $C_d(t)$ is greater than the intraclass distance. Therefore, it is possible to determine the quality of the reservoir before training.

2.3.2. Lyapunov coefficient

According to [54] and [55], the chaotic nature of the reservoir can be measured empirically by estimating the Lyapunov coefficient. If the value of the coefficient is positive, the system has a chaotic behavior and small changes in the input can present major changes in the states of the reservoir.

On the other hand, a negative Lyapunov coefficient means there is a stable network, where small differences in the input converge in reservoir states that tend to be an attractor. In [56] there is a method to calculate the Lyapunov coefficient for small data sets estimating the average distance for separation rate between two closest neighbors. The distance between states at time t is defined in equation (10).

$$d_j(t) = \|x_i(t) - x_j(t)\| \quad (10)$$

where k is the closest neighbor to $x_j(t)$. Assuming that the distance of states at time t is $d_j(t)$, the Lyapunov coefficient, γ , can be calculated for k number of elements to be classified using equation (11).

$$d_j(t) = \|u_i(t) - u_j(t)\| \quad (11)$$

where, $u_i(t)$ is the closest neighbor to $a(t)$ in the input vector. The Lyapunov coefficient is then defined by equation (12)

$$\gamma(t) = \frac{1}{k} \sum_{i=1}^P \ln \left(\frac{\|x_i(t) - x_j(t)\|}{\|u_i(t) - u_j(t)\|} \right) \quad (12)$$

where, $u_i(t)$ is the closest neighbor to $u_j(t)$ and $x_i(t)$ and $x_j(t)$ are the reservoir states resulting from the input vectors $u_i(t)$ and $u_j(t)$. According to [54], [57] and [58] it has been observed that the optimum performance of the reservoir takes place in the vicinity of chaos, where the Lyapunov coefficient is close to 0. In other words, the result of the ratio between the difference of states and difference of input vectors in a reservoir with an ideal response must comply with equation (13).

$$SR = \frac{\|x_i(t) - x_j(t)\|}{\|u_i(t) - u_j(t)\|} \cong 1 \quad (13)$$

2.3.3. Separation Ratio Graph

The concepts of reservoir separation and the Lyapunov coefficient are unified in a method called Separation Radius (SR) [42]. The SR is defined as the ratio between the distance of the input vectors and the distance between the states generated by these inputs, as shown in equation (13). An optimal reservoir meets $SR \cong 1$.

It is possible to graphically present the separation distance of the input vectors versus the separation distance of the corresponding reservoir states. To make this a graphical presentation, a set of input vectors $\{u_1(t), u_2(t), \dots, u_p(t)\}$ is used, and the states generated by these inputs $\{x_1(t), x_2(t), \dots, x_p(t)\}$. For each pair of states, i and $j \in [1, p]$, where p is the total number of patterns, the distance between the input vectors $\|u_i(t) - u_j(t)\|$, is plotted against the distance between the corresponding states $\|x_i(t) - x_j(t)\|$. As it is possible to observe in Fig. 2, there are 3 regions, the first one known as the chaotic zone presents us with very different responses from the reservoir for similar inputs, the area considered as attractors presents very similar responses for different inputs and finally there is the target area which has the desired behavior.

3. Echo State network - Genetic algorithm - separation Ratio Graph hybrid model

This section presents the proposal for a new hybrid neuro-evolutionary system, called Echo State Network - Genetic Algorithm - Separation Ratio Graph (ESN-GA-SRG), which optimizes the general parameters of an ESN using a genetic algorithm. In the case for forecasting time series that present a challenge due to their difficulty, the definition of the time input window is also part of the problems that must be solved, and which are considered in the proposed model. In addition, an analysis of the states generated by the reservoir is performed, using the Separation Ratio Graph method, to determine the appropriate configuration of the reservoir (connections and weights).

In addition to offering a greater understanding of the ESN design, the sensitivity analysis carried out in the previous works [59] allowed us to identify which are the parameters related to the reservoir that, when changed, generate greater variations in performance in the model. The most sensitive parameters were the number of units in the Reservoir, the percentage of reservoir connections and the weights of the reservoir connections.

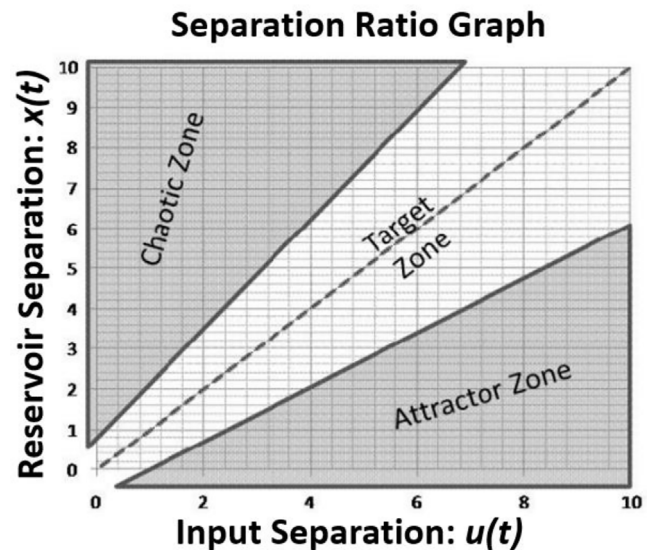


Fig. 2. Objective region and chaotic zones and attractors (adapted [42]).

The idea is to extend the SRG concept by [42] for classification, presented in the previous section, to ESNs applied to time series forecasting. The integration with the Genetic Algorithm, used for optimization, aims to facilitate the design and evaluation of the reservoir states. Thus, the following sections present an overview of the proposed hybrid model and detail its respective modules separately.

3.1. Overview

The proposed model is made up of 2 fundamental parts, initially there is a genetic algorithm that performs the task of optimizing the hyperparameters so that later the generation of a reservoir is carried out that will be evaluated using the SRG method once a coefficient is obtained of separation in the target area, the forecast of the time series is carried out.

As presented in Fig. 3, it begins in the upper left with the optimization of the hyperparameters, then the value of the forecast window is reused to determine the separation coefficient.

In the overall structure of the proposed model, two principal parts stand out:

- **NEURO-EVOLUTION:** This module is used to define the best configuration of global parameters of an ESN network for forecasting time series. For this task, a Genetic Algorithm (AG) is used, with n number of generations and the objective of determining the following parameters: Number of units in the reservoir (N), Connection Percentage (% C), Spectral Radius ρ (W), Scaling factor for input connection weights (sW_e) and Scaling factor for recurring connection weights sW_r . In addition to these parameters, the genetic algorithm also determines the size of the input window Jan , that is, the number of values of

the historical series that must be presented at the entrance of the ESN. As an aptitude function, an error metric is used to predict the validation set.

- **SRG:** After determining the best ESN network configuration by the Neuro-Evolution module, the SRG module will determine the best reservoir for the topology generated by Neuro-Evolution. To perform this reservoir selection procedure, n_r ESN networks are created with the best topology of the Neuro-Evolution module. For each of these n_r networks, the input patterns (which are previously paired using a neighborhood algorithm) are presented and the states generated by the n_r reservoirs' neurons are collected. Finally, the reservoir with the best SRG will be selected. This procedure is, therefore, a fine adjustment of the network, performing a local search performed for n_r initializations of the reservoir.

Next, in the next two sections, each of the modules that make up the proposed model will be presented in greater detail.

3.2. Neuro-Evolution module

As explained in the previous section, this module defines the best configuration of global hyperparameters using Genetic Algorithms [39] for the ESN network. The use of AG considers the intensity of the scale factors of the input connections and the recurring connections, which in the existing literature had not been included by optimization algorithms besides the window size, number of reservoir units, percentage of connection and radius spectral. The whole procedure is shown in Fig. 4.

Next sections present the methods used in each GA state that optimize ESN topology.

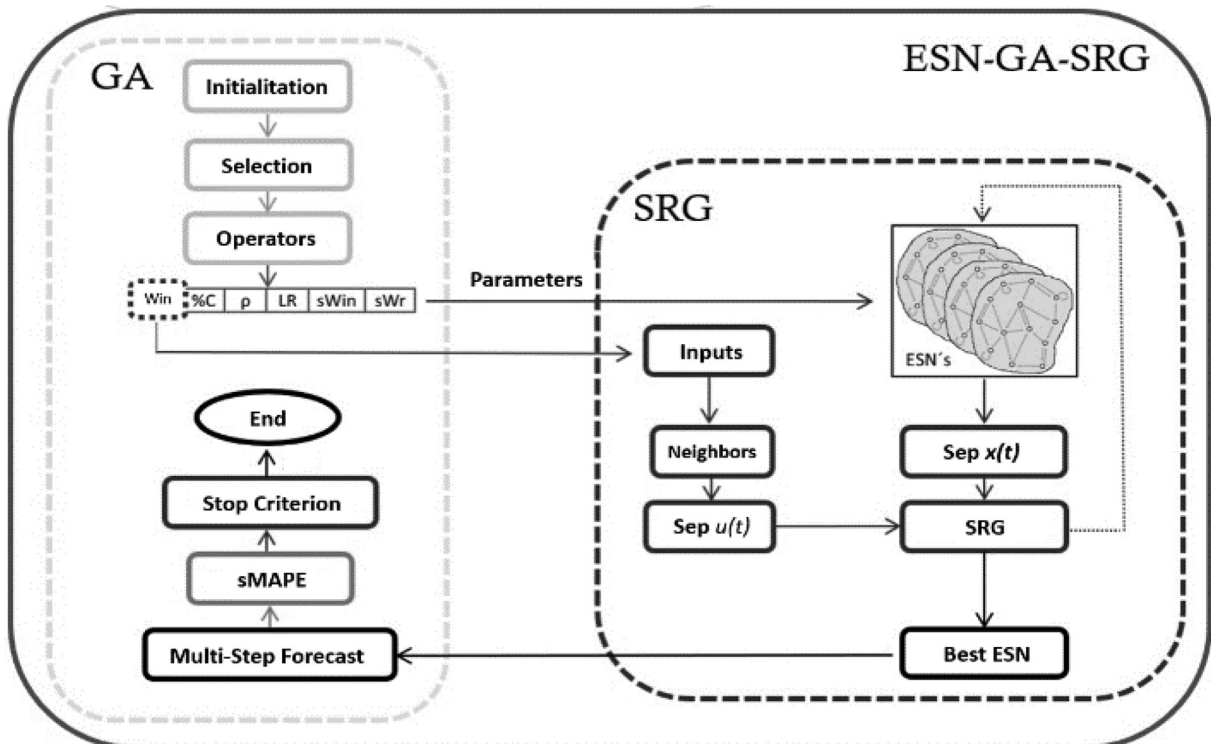


Fig. 3. Schematic diagram for the proposed model.

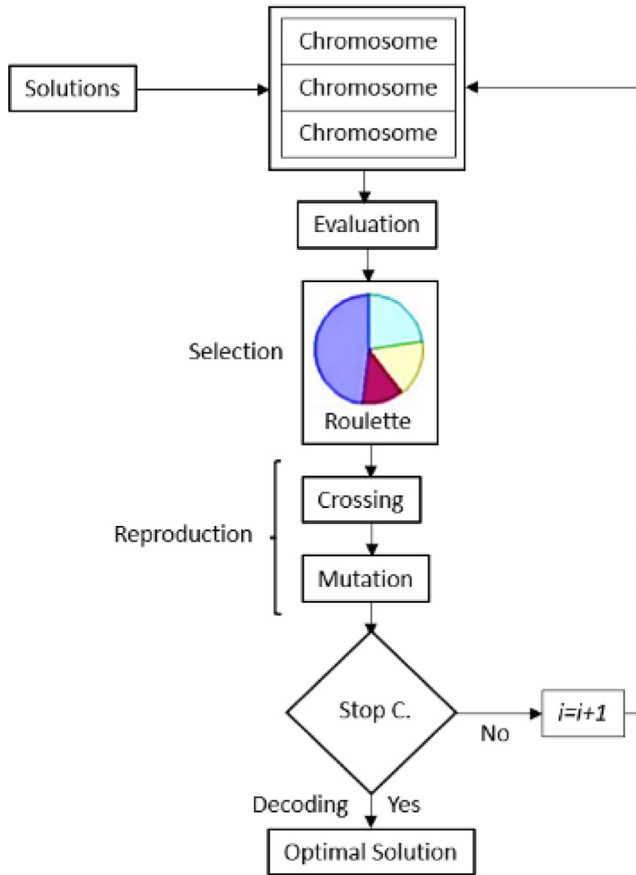


Fig. 4. Flowchart for the Genetic Algorithm.

3.2.1. Representation and coding

In the numerical representation, the chromosome size is defined by the number of parameters to be optimized. For the problem addressed, the first gene represents the size of the window, the second one is the number of units in the reservoir, the third gene is the percentage of connection, the fourth gene represents the Spectral Radius, and the last two genes represent the scale factor in the weights, the input connections and the weights of the recurring connections, respectively. Fig. 5 shows the composition of the chromosome, where Jan corresponds to the size of the Window, N is the number of units in the reservoir, $\%C$ corresponds to the connection percentage, $\rho(W)$ is Spectral Radius and sW^e and sW^{rr} are the factors of scale of the weights of the incoming connections and recurring connections, respectively.

3.2.2. Initialization and selection

The population initialization process generates solutions at random, like any evolutionary process, which is necessary for there to be diversity in the initial population, to cover the largest possible area of the search space. In the selection process, several methods can be applied, and according to [60], the roulette, which was used in this work. It has the advantage of having good results due to the high probability of selecting the most suitable individuals. So, the selection is carried out according to the probability that individuals have according to the initial assessment carried out. This way, it is

Jan	N	$\%C$	$\rho(W)$	sW^e	sW^{rr}
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Fig. 5. Chromosome Composition.

possible to guarantee greater diversity and have a set made up of significant samples.

3.2.3. Reproduction

The reproduction procedure, which allows the generation of new individuals imitating the natural process, is divided into three stages. The first one is mating, where pairs of individuals are selected; the second and third stages, known as crossing and mutation respectively, perform the recombination and modification of the pairs of selected individuals, generating new elements within the population. The genetic code is therefore passed on from parents to children at this stage. There are different methods, or operators, widely studied and applied to different problems in the literature [61]. The choice of these operators is related to the type of representation and coding.

The crossing operator selected in this work was the crossing of a point, which has advantages in conserving the chromosome information [62]. Fig. 6 shows an example of the procedure performed on this operator.

The crossing point (P.C.) is also selected at random, and then recombination is performed to create the children that contain part of the genetic code of the father (C1) and the mother (C2).

The adopted method for the mutation operator is the uniform, where a chromosome gene is randomly drawn, and it is changed to another random value within the limits defined for each variable. This allows slight changes in the chromosome and preserves the original information [63].

3.2.4. Fitness function

As fitness function, an error metric is used to evaluate the performance of the ESN neural network configured by the specific chromosome. In this work, the sMAPE obtained in the validation set was adopted, as it is the metric used in the case studies. However, any other error metric could be used, such as Mean Absolute Percentage Error (MAPE) or Root Mean Squared Error (RMSE). Equation (14) presents the error metric of the present work.

$$sMAPE_{Validation}(\%) = \frac{1}{N} \sum_{t=1}^N \frac{\|y_t - y'_t\|}{(y_t + y'_t)/2} * 100 \quad (14)$$

where, y_t is the actual value to be predicted, y'_t is the value predicted by the model and N is the number of valid samples.

3.3. Separation Ratio Graph module

The SRG model proposed in [42] was applied to classification tasks as presented in Section 2. In the present work, an extension of this model to multi-step time series forecasting tasks is proposed.

The use of this model allowed us to assess whether the states (Echoes) generated by the reservoir during the presentation of the patterns are interpreted correctly, i.e. similar entries (closest neighbors) generate similar responses (states). If the ratio of the distance of the responses to the distance of the inputs is close to 1, it can be interpreted that the reservoir has an ideal response. The reservoir shows undesirable behavior in two cases: the behavior is in the chaotic part (similar inputs-different responses) or behavior of an attractor (different inputs-same responses). Hence, this module is used to choose the reservoir with a behavior closer to the ideal, among the n possible initializations for the ESN with general optimized parameters generated by the Neuro-Evolution module.

3.3.1. Separation Ratio estimate

As presented in Section 2, it is necessary to carry out a grouping in pairs for the entire set of patterns, to select pairs of closest

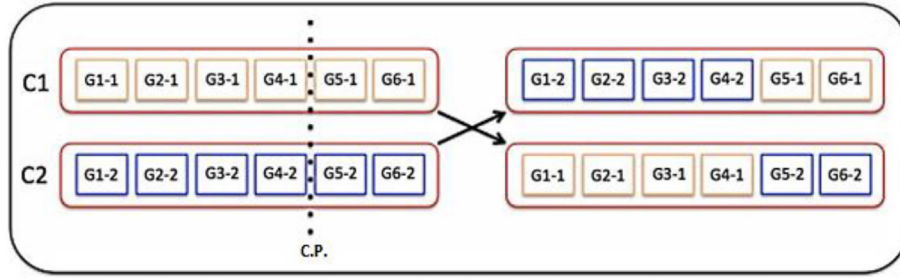


Fig. 6. Example of crossing a point.

neighbors and present them to the reservoir. After this process, the states resulting from each pair of patterns presented are collected. In the proposed model the grouping is performed using the k -nearest neighbors' algorithm – knn , where the Euclidean distance is used as a proximity metric. Equation (15) provides an example for two patterns i and j .

$$d_{ij}(u_i, u_j) = d_{j,i}(u_j, u_i) = \sqrt{(u_j, u_i)^2} \quad (15)$$

The k value used in the model is equal to 1, since only 1 nearest neighbor is sought for each pattern. Thus, for each pattern, the distances are obtained for all available patterns, selecting the pattern that provides the lowest distance value. Therefore, from a base with P training patterns, a set of P pairs g_l , $l = \{1, \dots, P\}$ is obtained, associating each input pattern u_i with its closest neighbor u_j . Additionally, for each input pattern of a g_i pair, the resulting states of the reservoir (x_i and x_j) are calculated. The state generation procedure must comply with the condition that all states X are generated with a pattern P starting from the same place, that is, the immediately previous state is not considered historical to generate the current state. Equation (16) is used to identify the SRG coefficient for each generated reservoir. Then, the selection of the best reservoir is performed as the linear regression obtained.

$$SRG = \sum_{p=1}^N |||dif_p(u)|| - ||dif_p(x)|||| \quad (16)$$

where $dif_p(u)$ is the distance between the input values of two patterns that make up the pair p ; $dif_p(x)$ is the distance between the two states generated in the reservoir by the patterns that make up the pair p ; and N is the total number of pattern pairs generated by the clustering and neighborhood algorithm.

3.3.2. Reservoir choice

To interpret the results obtained by each reservoir, a linear regression is performed using the distances of the input vectors and the distances of the generated states. The best configuration is obtained from the line generated after the regression (see Equation (17)), where the parameters that make up the line equation are evaluated:

$$y = mx + b \quad (17)$$

Where m is the slope and b is the intersection point with the y axis. For the reservoir to show an ideal behavior, the line resulting from the linear regression must be on the dotted line, as shown in Fig. 7. That is, $m = 1$ and $b = 0$. Consequently, the criterion for selecting the best reservoir is through the regression that results in m nearest 1 and b closest to 0.

Additionally, the correlation coefficient R is calculated using equation (18), which determines the linear relationship between the distance of the input vectors and the distance of the generated states as shown in Fig. 7.

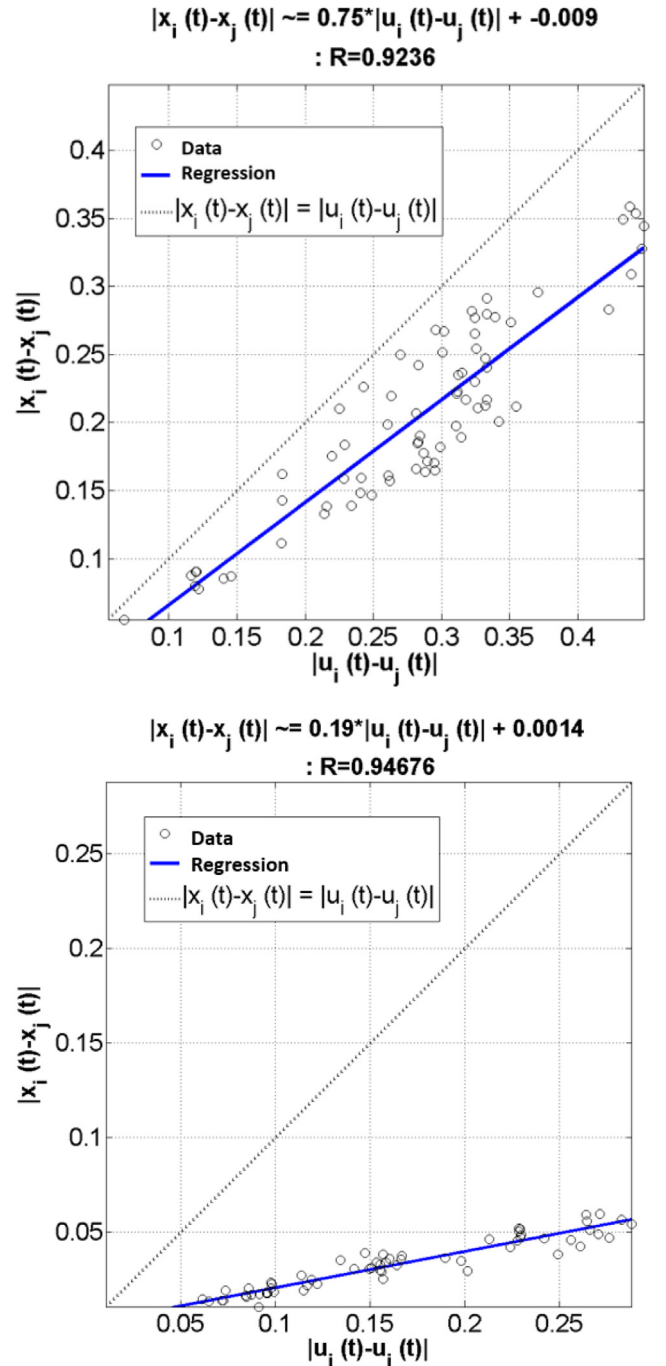


Fig. 7. Examples of regressions obtained in different evaluations to choose the best reservoir.

$$R = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{(\sum (x_i - \bar{x})^2)(\sum (y_i - \bar{y})^2)}} \quad (18)$$

This section presents the various experiments and tests performed with the proposed ESN-GA-SRG model, comparing its results with: traditional ESN models, other ESN models with parameter optimization by genetic algorithms and traditional statistical models [65] and based on forecasting committees, described in [66], for multi-step forecasting. All models were evaluated with the sMAPE metric for performance analysis in multi-step prediction on out-of-sample data sets.

3.4. Description of databases

This work used, as a case study, the reduced version (11 series) of the NN3 Forecasting Competition for Neural Networks & Computational Intelligence, and 36 series of the M3 Forecasting Competition, selected according to: sampling, asymmetry, seasonality and stationarity. The choice of these benchmark bases is based on the following facts:

1. Official multi-step forecast competition, world-class, with the participation of the main research institutes.
2. Composed of different time series, which allow for a complete assessment of the performance of models proposed for such purposes.
3. Access to the list of results of the competition for all the series proposed.

Tables 1 to 4 show information on the series used, such as the size of the total base, the amount of data used for training and the size of the validation and test sets, which were defined according to the competition regulations. In addition, the tables indicate the asymmetry coefficient of each series which calculates the asymmetric variability of these series, where values close to zero indicate a symmetric frequency distribution, which can be interpreted as series easier to predict. The asymmetry coefficient can be obtained through equation (19).

$$Y = \frac{E[(X - \bar{X})^3]}{\sigma^3} \quad (19)$$

where $E[(X - \bar{X})^3]$ is the third central moment of a random variable X and $\bar{X}$ is the expected value of X , known as μ_3 and σ is the standard deviation.

Table 1
General information of the 11 NN3 series.

Series	Total Data	Train Set	Validation Set	Test Set	Asymmetry
101	144	126	18	18	-0.0985
102	144	126	18	18	-0.1476
103	144	126	18	18	1.3085
104	133	115	18	18	-0.7074
105	144	126	18	18	-0.1508
106	144	126	18	18	0.2233
107	144	126	18	18	0.2815
108	133	115	18	18	1.7009
109	141	123	18	18	1.3751
110	144	126	18	18	4.4008
111	144	126	18	18	0.8792

Table 5 shows the limits used in the chromosome configuration of the genetic algorithm used to configure the ESN base model, as well as the resolution used for each parameter.

The results obtained in the experiments carried out with the proposed model ESN-GA-SRG and their comparison with the different models in the literature are presented below.

3.5. Forecast result for NN3 set

For each of the 11 series, the final configuration parameters of the ESN-GA-SRG model are presented, the results of the out-of-sample set forecast (test) for the ESN, ESN-GA and ESN-GA-SRG models, a evolution of sMAPE over the forecast horizon (18 steps forward), and the linear regression obtained for the reservoir before and after using the SRG method.

Table 6 shows the average of the sMAPE for the test sets of the 11 series of the competition, and in each of the models generated the results obtained by other statistical models such as: Multiplicative Holt-Winters (HW), Harmonic Regression (REG), Classical decomposition (DEC), the HW / REG / DEC Experiment (Method: AFTER for expansive window & static weights) and Box & Jenkins (BJ). These results were published in [63], where the Neural Expert Weighting (NEW) model was proposed, which was also included in the comparative table below. It is possible to observe that the proposed ESN/GA/SRG model obtained the lowest value with the sMAPE metric for the test group.

Table 7 shows the final configuration parameters of the ESN networks for the NN3 set series after ESN/GA/SRG model run.

It can be concluded, from the evaluation of table 8, that the use of the ESN-GA-SRG model improved the response of the other models used in 10 of the 11 series, which make up the case study. Additionally, the interpretation of the reservoir behavior during the predictions made was clearer, that is, the results of the states helped understand more fully the behavior of the ESN networks.

Table 8 shows the sMAPE results for each of the 11 series individually. As can be seen, the ESN-GA-SRG model managed to be the winner in 4 series (2, 3, 6 y 8) and in the average sMAPE of the 11 series.

Table 9 shows the results of the first 10 places in the NN3-Forecasting competition, allowing us to observe that the proposed model would have achieved the best position. It is important to note that the runner-up (Ilies, Jäger, Kosuchinas, Rincon, Sakenas, Vaskevicius), who obtained a sMAPE of 14.26, used ESN networks with a methodology where the forecasts [64] made by different ESN networks were weighted for the final forecast.

3.5.1. Result of statistical significance tests for the NN3 series

In order to statistically evaluate the results obtained in multi-step forecasts for each of the 11 series, parametric and non-

Table 2

General information of the 12 M3 series with Annual Sampling.

Series	Total Data	Train Set	Validation Set	Test Set	Asymmetry
33	20	8	6	6	-1,4141
50	20	8	6	6	-0,6796
70	20	8	6	6	-1,2893
105	20	8	6	6	0,6033
119	20	8	6	6	7,7543
162	21	9	6	6	-0,6440
171	47	35	6	6	-0,0091
200	43	31	6	6	104,525
351	20	8	6	6	-1,3609
564	25	13	6	6	2,8818
599	25	13	6	6	-1,1126
642	36	24	6	6	-0,9847

Table 3

General information on the 12 M3 series with Quarterly Sampling.

Series	Total Data	Train Set	Validation Set	Test Set	Asymmetry
671	44	28	8	8	0,0443
677	44	28	8	8	-1,1469
692	44	28	8	8	-0,0068
713	44	28	8	8	0,1067
730	43	27	8	8	-1,0847
758	44	28	8	8	-0,8037
773	44	28	8	8	-0,9884
829	44	28	8	8	-0,3549
847	43	27	8	8	23,3153
854	59	43	8	8	-0,6452
865	64	48	8	8	-1,0614
896	56	40	8	8	-0,7616

Table 4

General information of the 12 M3 series with Monthly Sampling.

Series	Total Data	Train Set	Validation Set	Test Set	Asymmetry
1430	69	33	18	18	1,0424
1460	69	33	18	18	2,1892
1725	126	110	18	18	0,2113
1900	140	124	18	18	0,0985
1906	134	118	18	18	-0,8738
1912	144	128	18	18	-1,0965
1996	144	128	18	18	-0,2653
2001	144	128	18	18	-0,4481
2013	144	128	18	18	-0,9488
2016	144	128	18	18	-0,6650
2089	144	128	18	18	-1,0064
2579	134	118	18	18	-0,9716

Table 5

Limits of chromosome parameters.

Parameter	Symbol	Min. Value	Max. Value	Accuracy
Window	Jan	1	36	1
Number of Units	$\#U$	100	2000	1
Connection Percentage	$\%C$	20	100	1
Spectral Radius	$r(W)$	0.1	1	0.1
Scale Factor in the Input Connections weights	sW^e	0	1	0.1
Scale Factor in the Input Connections weights	sW^{rr}	0	1	0.1

Table 6
Comparative results for the NN3 series.

Model	sMAPE_Test
ESN	15,95
ESN-GA	14,55
ESN-GA-SRG	11,51
HW	15,08
REG	16,68
DEC	16,25
HW/REG/DEC	13,71
BJ	15,03
NEW	13,84

parametric tests presented below were used. More information and discussions regarding the tests performed can be found in [67].

Test	Rating
T-Student	Parametric
Sync	Non- Parametric
Wilcoxon (Signed Rank)	Non- Parametric
Friedman	Non- Parametric
Post-Hoc	Non- Parametric

The value established for the level of significance, identified as p-value, was 5%, and the comparison of the different models, which stood out the most in the results presented (table 8), is performed using the accumulated sMAPE for each step of the forecast horizon. Tables 10 to 12 show the results for the t-student, Sync and Wilcoxon tests, where it is possible to verify that the proposed model

managed to have statistically significant differences in most cases, when the comparison is made pairwise.

Table 13 shows the result obtained for the Friedman test, where the six selected models are compared simultaneously. It is possible to observe that the proposed model managed to obtain the first position in the comparison.

Table 14 shows the result obtained for the Post-Hoc test, where again all models are compared simultaneously, but this time, the significance level is obtained for all selected models. Again, it is possible to observe that the proposed model managed to obtain statistically significant differences for 4 of the 6 models.

Regarding the preprocessing, in all the time series were evaluated in the coefficient of asymmetry, the seasonality was removed, and the input values were normalized to avoid scattered values.

Through statistical validation it was possible to verify that the responses obtained in the forecast by the proposed model deliver that in 4 of the 6 comparison cases they are better, this was done for the Post Hoc and Wilcoxon tests [65].

Table 9
Results of the first 10 places in the NN3 competition.

Position	Model	sMAPE
	ESN-GA-SRG	11,51
1	Yan	13,68
2	Ilies, Jäger, Kosuchinas, Rincon, Sakenas, Vaskevcius	14,26
3	Chan, Yao	14,46
4	Yousefi, Miromeni, Lucas	14,49
5	Ahmed, Atiya, Gayar, El-Shishiny	14,52
6	Flores, Anaya, Ramirez, Morales	15,00
7	Adeodato, Vasconcelos, Arnaud, Chunha, Monteiro	15,10
8	Luna, Soares, Ballini	15,35
9	Theodosiou, Swamy	16,19
10	Hwang, Song, Kasabov	16,31

Table 7
Final configuration parameters for the NN3 series.

Parameter Series	Jan	#U	%C	r(W)	sW ^e	sW ^{rr}
101	14	381	30	0.6	0.3	0.7
102	12	412	52	0.5	0.5	0.4
103	12	365	45	0.4	0.7	0.4
104	9	637	27	0.6	0.5	0.4
105	13	271	65	0.7	0.3	0.8
106	8	368	72	0.8	0.6	0.6
107	10	405	89	0.4	0.7	0.2
108	9	195	34	0.7	0.4	0.5
109	6	748	23	0.8	0.7	0.1
110	12	437	60	0.4	0.5	0.6
111	11	331	49	0.9	0.3	0.8

Table 8
Results for each of the NN3 series.

Series	HW	REG	DEC	HW /REG /DEC	BJ	NEW	ESN	ESN-GA	ESN-GA-SRG
1	1,80	2,67	2,61	1,95	1,99	2,64	3,71	3,15	2,82
2	30,63	24,55	29,01	24,77	12,64	22,05	14,82	13,14	9,52
3	29,60	33,81	29,16	33,81	44,01	34,91	35,21	27,16	27,16
4	6,11	9,37	13,78	8,13	4,96	6,41	17,24	12,32	10,87
5	1,53	10,22	10,66	1,62	2,67	8,28	4,03	3,32	2,52
6	5,38	3,62	4,10	4,08	5,31	6,07	3,65	3,28	2,19
7	5,10	3,48	3,29	3,02	5,87	3,94	6,49	6,56	4,14
8	36,29	29,54	29,62	30,35	29,58	25,83	28,14	29,69	20,43
9	7,77	26,88	16,69	6,71	7,02	8,62	12,25	11,63	8,07
10	30,48	27,55	28,62	27,25	33,61	23,02	35,16	34,44	27,79
11	11,14	11,79	11,23	11,79	17,75	10,49	14,72	15,36	11,10

Table 10Results of the *t*-student test for the NN3 series.

t-student Model	p-value	Inferior Lim.	Superior Lim.	Conclusion
ESN	4,95e-13	-4,9258	-3,9252	ESN + GA + SRG is better
ESN + GA	7,05e-10	-2,6911	-1,9020	ESN + GA + SRG is better
NEW	0,2132	-1,6480	0,3955	Equivalent
TRAD	4,09e-8	-3,5469	-2,2412	ESN + GA + SRG is better
HW	0,0522	-1,9417	0,0103	Equivalent
BJ	5,86e-8	-4,5670	-2,8510	ESN + GA + SRG is better

Table 11

Sync test results for the NN3 series.

Sync Test Model	p-value	Inferior Lim.	Superior Lim.	Conclusion
ESN	7,62e-6	-5,0435	-4,0349	ESN + GA + SRG is better
ESN + GA	7,62e-6	-2,8911	-1,7094	ESN + GA + SRG is better
NEW	0,0309	-1,6666	-0,0227	ESN + GA + SRG is better
TRAD	1,44e-4	-3,4366	-2,7927	ESN + GA + SRG is better
HW	0,2379	-2,2997	0,6573	Equivalent
BJ	1,44e-4	-4,1143	-3,2727	ESN + GA + SRG is better

Table 12

Wilcoxon test results for the NN3 series.

Wilcoxon Test Model	p-value	Inferior Lim.	Superior Lim.	Conclusion
ESN	1,80e-4	-4,9358	-4,1027	ESN + GA + SRG is better
ESN + GA	1,80e-4	-2,7995	-1,8844	ESN + GA + SRG is better
NEW	0,0407	-1,6666	-0,0016	ESN + GA + SRG is better
TRAD	2,53e-4	-3,5662	-2,6785	ESN + GA + SRG is better
HW	0,0451	-2,1617	0,0187	ESN + GA + SRG is better
BJ	2,14e-4	-4,6085	-3,2181	ESN + GA + SRG is better

Table 13

Results for the Friedman test for the NN3 series.

Friedman Test							
Model	ESN + GA + SRG	NEW	Traditional	HW	BJ	ESN	ESN + GA
h = 1	5	7	6	1	4	2	3
h = 2	3	7	4	1	5	2	6
h = 3	3	7	5	2	4	1	6
h = 4	3	7	4	2	5	1	6
h = 5	2	7	4	3	5	1	6
h = 6	2	7	4	3	5	1	6
h = 7	1	7	4	2	6	3	5
h = 8	1	7	4	3	5	2	6
h = 9	1	6	4	3	5	2	7
h = 10	1	6	4	3	5	2	7
h = 11	1	6	4	3	5	2	7
h = 12	1	6	4	3	5	2	7
h = 13	1	6	4	2	5	3	7
h = 14	1	6	4	2	5	3	7
h = 15	1	7	5	2	4	3	6
h = 16	1	7	3	2	4	5	6
h = 17	1	7	3	2	4	5	6
h = 18	1	7	3	2	4	6	5
Sum	30	120	73	41	85	46	109
Average	1,66	6,66	4,05	2,27	4,72	2,55	6,05
Position	1	7	4	2	5	3	6

3.6. Forecast result for M3 set

The results obtained in the experiments carried out with the proposed model ESN-GA-SRG and their comparison with the six best positioned models for each set of selected series of the M3 competition are presented next.

Initially, the global base available was separated into three groups according to sampling: annual, quarterly, and monthly with forecast horizons of 6, 8 and 18 steps ahead respectively. Within each of these 3 groups (annual, quarterly, and monthly), 12 series were selected for each group according to characteristics such as asymmetry, stationarity and seasonality, as shown in [tables 15 to 17](#).

Table 14

Results for the Post-Hoc test for the NN3 series.

Model	Model	Inferior Lim.	Average Dif.	Superior Lim.	p-Value	Conclusion
ESN + GA + SRG	NEW	−7,1230	−5,0000	−2,8770	0,0000	ESN + GA + SRG is better
ESN + GA + SRG	Traditional	−4,5119	−2,3889	−0,2659	0,0158	ESN + GA + SRG is better
ESN + GA + SRG	HW	−2,7341	−0,6111	1,5119	0,9797	Equivalent
ESN + GA + SRG	BJ	−5,1786	−3,0556	−0,9325	0,0004	ESN + GA + SRG is better
ESN + GA + SRG	ESN	−3,0119	−0,8889	1,2341	0,8808	Equivalent
ESN + GA + SRG	ESN + GA	−6,5119	−4,3889	−2,2659	0,0000	ESN + GA + SRG is better
NEW	Traditional	0,4881	2,6111	4,7341	0,0053	Equivalent
NEW	HW	2,2659	4,3889	6,5119	0,0000	Equivalent
NEW	BJ	−0,1786	1,9444	4,0675	0,0981	Equivalent
NEW	ESN	1,9881	4,1111	6,2341	0,0000	Equivalent
NEW	ESN + GA	−1,5119	0,6111	2,7341	0,9797	Equivalent
Traditional	HW	−0,3453	1,7778	3,9008	0,1707	Equivalent
Traditional	BJ	−2,7897	−0,6667	1,4564	0,9685	Equivalent
Traditional	ESN	−0,6230	1,5000	3,6230	0,3627	Equivalent
Traditional	ESN + GA	−4,1230	−2,0000	0,1230	0,0802	Equivalent
HW	BJ	−4,5675	−2,4444	−0,3214	0,0122	HW is better
HW	ESN	−2,4008	−0,2778	1,8453	0,9997	Equivalent
HW	ESN + GA	−5,9008	−3,7778	−1,6547	0,0000	HW is better
BJ	ESN	0,0436	2,1667	4,2897	0,0419	Equivalent
BJ	ESN + GA	−3,4564	−1,3333	0,7897	0,5128	Equivalent
ESN	ESN + GA	−5,6230	−3,5000	−1,3770	0,0000	ESN is better

Table 15

M3 Series selected for the annual sampling group.

Annual	Asymmetry			
Stationarity	Seasonality	Low	Medium	High
No	No	M3_171	M3_105	M3_119
No	Yes	M3_050	M3_033	M3_564
Yes	No	M3_162	M3_642	M3_070
Yes	Yes	M3_599	M3_351	M3_200

Table 16

M3 Series selected for the quarterly sampling group.

Quarterly	Asymmetry			
Stationarity	Seasonality	Low	Medium	High
No	No	M3_896	M3_730	M3_847
No	Yes	M3_758	M3_773	M3_865
Yes	No	M3_692	M3_713	M3_854
Yes	Yes	M3_671	M3_829	M3_677

Table 17

M3 Series selected for the monthly sampling group.

Monthly	Asymmetry			
Stationarity	Seasonality	Low	Medium	High
No	No	M3_1996	M3_1430	M3_1460
No	Yes	M3_1725	M3_2089	M3_1912
Yes	No	M3_2001	M3_2016	M3_2579
Yes	Yes	M3_1900	M3_1906	M3_2013

Table 18

Results for the M3 annual series Post-Hoc test.

Annual Model	sMAPE
ForecastPro	17,05
Flors_Pearc2	17,20
AutoBox2	17,38
Flors_Pearc1	17,38
SMARTCFS	17,39
THETA	17,40
ESN-GA-SRG	8,88

3.6.1. Annual series results

Table 18 shows the comparative average results for the 12 series selected for the annual sampling group. It is possible to observe that, according to the sMAPE metric, the proposed model achieved the smallest error in the multi-step forecast, in the case of ForecastPro, which is a computational platform, from which several models were used, such as Exponential Smoothing / Box & Jenkins / Poisson.

Table 19 shows the results of sMAPE for each of the 12 series, allowing us to compare the results obtained in the 12 series individually.

Table 19

Comparative results for each annual M3 series.

sMAPE Series	Forecast Pro	Flors Pearc2	Auto Box2	Flors Pearc1	SMARTFCS	THETA	ESN + GA+ SRG
33	32,80	12,56	34,91	7,48	18,96	12,55	6,26
50	13,73	12,22	15,23	12,22	26,28	21,66	7,32
70	9,59	16,20	9,92	10,48	8,41	8,71	8,17
105	21,54	11,17	7,55	15,66	15,21	18,91	5,52
119	31,66	29,53	24,88	32,03	29,66	31,34	26,41
162	29,31	19,24	16,88	16,50	30,82	21,06	14,55
171	3,14	31,94	17,17	38,46	17,58	26,09	3,10
200	14,85	12,88	33,14	14,77	12,79	13,88	11,24
351	17,38	22,06	20,56	22,06	12,32	22,59	8,46
564	12,31	11,31	13,31	11,31	8,29	9,48	8,42
599	5,28	13,76	4,01	14,12	15,18	8,92	2,14
642	12,96	13,46	10,98	13,46	13,25	13,65	5,09

Table 20

Parameters of the ESN-GA-SRG model for the annual M3 series.

Series	Parameter					
	Jan	#U	%C	$\rho(W)$	sW^e	sW^{rr}
033	6	240	90	1	0.6	0.6
050	6	110	20	0.7	0.7	0.2
070	5	120	20	0.9	0.6	0.3
105	4	186	23	0.8	0.5	0.4
119	5	287	90	0.2	0.8	0.3
162	6	308	30	0.7	0.1	0.9
171	6	213	25	0.6	0.7	0.4
200	7	279	21	0.6	0.6	0.5
351	4	301	22	0.4	0.3	0.5
564	3	195	40	0.2	0.2	0.5
599	5	173	10	0.6	0.5	0.6
642	5	342	80	0.6	0.7	0.4

Table 21Results of the *t-student* test for the M3 series with annual sampling.

t-student Model	<i>p-value</i>	Inferior Lim.	Superior Lim.	Conclusion
ForecastPro	9,58e-04	-8,8440	-4,0593	ESN + GA + SRG is better
Flors_Pearc2	0,0028	-8,9778	-3,2292	ESN + GA + SRG is better
Autobox2	0,0312	-9,1205	-0,6573	ESN + GA + SRG is better
Flors_Pearc1	0,0054	-8,9208	-2,6009	ESN + GA + SRG is better
SMARTFCS	0,0070	-9,1987	-2,4160	ESN + GA + SRG is better
THETA	0,0349	-9,0218	-0,5013	ESN + GA + SRG is better

Table 22

Sync test results for M3 series with annual sampling.

Model	<i>p-value</i>	Inferior Lim.	Superior Lim.	Conclusion
ForecastPro	0,0313	-8,3458	-2,7967	ESN + GA + SRG is better
Flors_Pearc2	0,0313	-8,4055	-1,7118	ESN + GA + SRG is better
Autobox2	0,2188	-8,4942	1,1614	Equivalent
Flors_Pearc1	0,0313	-8,4968	-1,2894	ESN + GA + SRG is better
SMARTFCS	0,0313	-8,5349	-0,9250	ESN + GA + SRG is better
THETA	0,2188	-8,5150	0,8146	Equivalent

Table 23

Wilcoxon test results for the M3 series with annual sampling.

Model	<i>p-value</i>	Inferior Lim.	Superior Lim.	Conclusion
ForecastPro	0,0313	-8,3458	-2,7967	ESN + GA + SRG is better
Flors_Pearc2	0,0313	-8,4055	-1,7118	ESN + GA + SRG is better
Autobox2	0,0938	-8,4942	1,1614	Equivalent
Flors_Pearc1	0,0313	-8,4968	-1,2894	ESN + GA + SRG is better
SMARTFCS	0,0313	-8,5349	-0,9250	ESN + GA + SRG is better
THETA	0,0938	-8,5150	0,8146	Equivalent

Table 24

Results for the Friedman test for the annual M3 series.

Model	ESN + GA+ SRG	Forecast Pro	Flors Pearc2	Auto Box2	Flors Pearc1	SMARTFCS	THETA
h = 1	2	7	6	1	5	4	3
h = 2	2	7	6	4	5	3	1
h = 3	1	7	6	3	4	5	2
h = 4	1	7	5	4	3	6	2
h = 5	1	3	5	2	6	7	4
h = 6	1	2	3	4	5	6	7
Sum	8	33	31	18	28	31	19
Average	1,3333	5,5000	5,1667	3,0000	4,6667	5,1667	3,1667
Position	1	7	5	2	4	5	3

Table 25

Results for the Post-Hoc test for the annual M3 series.

Model	Model	Inferior Lim.	Average Dif.	Superior Lim.	p-Value	Conclusion
ESN + GA + SRG	ForecastPro	-7,843	-4,166	-0,489	0,0147	ESN + GA + SRG is better
ESN + GA + SRG	Flors_Pearc2	-7,510	-3,833	-0,156	0,0345	ESN + GA + SRG is better
ESN + GA + SRG	Autobox2	-5,343	-1,666	2,010	0,8349	Equivalent
ESN + GA + SRG	Flors_Pearc1	-7,010	-3,333	0,343	0,1052	Equivalent
ESN + GA + SRG	SMARTFCS	-7,510	-3,833	-0,156	0,0345	ESN + GA + SRG is better
ESN + GA + SRG	THETA	-5,510	-1,833	1,843	0,7628	Equivalent
ForecastPro	Flors_Pearc2	-3,3438	0,3333	4,0105	0,9999	Equivalent
ForecastPro	Autobox2	-1,1772	2,5000	6,1772	0,4116	Equivalent
ForecastPro	Flors_Pearc1	-2,8438	0,8333	4,5105	0,9942	Equivalent
ForecastPro	SMARTFCS	-3,3438	0,3333	4,0105	0,9999	Equivalent
ForecastPro	THETA	-1,3438	2,3333	6,0105	0,4998	Equivalent
Flors_Pearc2	Autobox2	-1,5105	2,1666	5,8438	0,5908	Equivalent
Flors_Pearc2	Flors_Pearc1	-3,1772	0,5000	4,1772	0,9996	Equivalent
Flors_Pearc2	SMARTFCS	-3,6772	0,0000	3,6772	1,0000	Equivalent
Flors_Pearc2	THETA	-1,6772	2,0000	5,6772	0,6800	Equivalent
Autobox2	Flors_Pearc1	-5,3438	-1,6666	2,0105	0,8349	Equivalent
Autobox2	SMARTFCS	-5,8438	-2,1666	1,5105	0,5908	Equivalent
Autobox2	THETA	-3,8438	-0,1666	3,5105	1,0000	Equivalent
Flors_Pearc1	SMARTFCS	-4,1772	-0,5000	3,1772	0,9997	Equivalent
Flors_Pearc1	THETA	-2,1772	1,5000	5,1772	0,8934	Equivalent
SMARTFCS	THETA	-1,6772	2,0000	5,6772	0,6801	Equivalent

Table 26

Comparative results for the quarterly M3 series.

Quarterly Model	sMAPE
ForecastrPro	4,42
ForcX	5,06
Holt	5,38
THETA	5,64
B_J_Auto	5,73
Arama	5,96
ESN-GA-SRG	3,86

Table 27

Comparative results for each quarterly M3 series.

sMAPE Series	Forecast Pro	ForcX	Holt	THETA	B_J_Auto	Arama	ESN-GA-SRG
671	3,91	1,84	3,03	2,83	8,51	3,59	1,42
677	1,09	5,31	4,06	4,14	1,09	1,78	1,09
692	4,16	5,55	3,78	3,79	3,65	5,94	3,35
713	4,39	5,71	4,08	5,49	3,96	3,88	3,46
730	3,71	3,72	5,30	6,66	6,04	5,47	3,28
758	3,68	3,40	2,60	6,56	3,47	9,08	2,17
773	8,79	8,65	17,83	7,09	8,62	9,89	6,48
829	3,55	7,53	3,85	10,84	6,98	3,09	5,74
847	11,72	10,57	10,24	10,69	16,95	16,95	11,04
854	3,20	2,93	3,06	2,87	2,89	2,70	2,70
865	2,59	2,78	4,02	4,13	4,01	6,13	3,49
896	2,29	2,85	2,83	2,65	2,66	3,14	2,16

Table 20 shows the configuration parameters of the ESN networks for the series that make up the M3 Annual set.

The tests of statistical significance applied to the results of the M3 series, whether annual, quarterly or monthly, are the same as those applied to the NN3 series. Next, the results obtained are presented (tables 21 to 23). As in the case of the NN3 competition, the value established for the level of significance, identified as *p-value*, was 5%, and the comparison of the different statistical models is performed using the accumulated sMAPE for each step of the forecast horizon.

Table 24 shows the result obtained for the Friedman test, where the six selected statistical models are compared simultaneously. It

Table 28

Parameters of the ESN-GA-SRG model for the M3 quarterly series.

Series	Parameter					
	Jan	#U	%C	$\rho(W)$	sW^e	sW^{rr}
671	5	251	26	0.8	0.8	0.1
677	4	358	34	0.7	0.6	0.4
692	5	259	33	0.5	0.6	0.6
713	4	430	36	1	0.2	0.8
730	7	327	21	0.4	0.5	0.5
758	5	198	30	0.9	0.9	0.3
773	3	429	50	0.4	0.9	0.1
829	9	346	43	0.9	0.6	0.5
847	4	216	20	0.3	0.1	0.8
854	5	336	20	1	0.9	0.3
865	9	264	36	0.8	0.9	0.2
896	11	256	24	0.7	0.3	0.5

Table 29

Results of the t-student test for the M3 series with quarterly sampling.

Model	<i>p-value</i>	Inferior Lim.	Superior Lim.	Conclusion
ForecastPro	0,2736	−0,1876	0,5664	Equivalent
ForcX	0,7135	−0,5988	0,4321	Equivalent
Holt	0,8479	−0,8768	1,0379	Equivalent
THETA	0,0893	−1,2339	0,1119	Equivalent
B_J_Auto	0,1861	−1,4325	0,3361	Equivalent
Arama	0,1561	−1,7219	0,3379	Equivalent

Table 30

Sync test results for M3 series with quarterly sampling.

Model	<i>p-value</i>	Inferior Lim.	Superior Lim.	Conclusion
ForecastPro	0,7266	−0,5633	0,9890	Equivalent
ForcX	0,7266	−1,2028	0,6872	Equivalent
Holt	1	−1,5242	2,0962	Equivalent
THETA	0,0893	−1,2339	0,1119	Equivalent
B_J_Auto	0,1861	−1,4325	0,3361	Equivalent
Arama	0,2891	−2,1050	1,7559	Equivalent

Table 31

Wilcoxon test results for the M3 series with quarterly sampling.

Model	<i>p-value</i>	Inferior Lim.	Superior Lim.	Conclusion
ForecastPro	0,3125	−0,2388	0,6298	Equivalent
ForcX	0,9453	−0,6214	0,4356	Equivalent
Holt	1	−0,8786	1,1535	Equivalent
THETA	0,0781	−1,1852	0,2532	Equivalent
B_J_Auto	0,1484	−1,3691	0,5735	Equivalent
Arama	0,1953	−1,6631	0,4446	Equivalent

Table 32

Results for the Friedman test for the M3 quarterly series.

Model	ESN + GA+ SRG	Forecast Pro	ForcX	Holt	THETA	B_J_auto	Arama
h = 1	7	4	6	1	5	3	2
h = 2	5	2	3	1	7	6	4
h = 3	4	2	3	1	6	7	5
h = 4	4	3	1	2	5	6	7
h = 5	3	1	2	4	5	6	7
h = 6	1	2	3	4	5	6	7
h = 7	1	2	3	4	5	6	7
h = 8	1	2	3	4	5	6	7
Sum	26	18	24	21	43	46	46
Average	3,2500	2,2500	3,0000	2,6250	5,3750	5,7500	5,7500
Position	4	1	3	2	5	6	6

Table 33

Results for the Post-Hoc test for the M3 quarterly series.

Model	Model	Inferior Lim.	Average Dif.	Superior Lim.	p-Value	Conclusion
ESN + GA + SRG	ForecastPro	-2,1845	1,0000	4,1845	0,9685	Equivalent
ESN + GA + SRG	ForcX	-2,9345	0,2500	3,4345	1,0000	Equivalent
ESN + GA + SRG	Holt	-2,5595	0,6250	3,8095	0,9974	Equivalent
ESN + GA + SRG	THETA	-5,3095	-2,1250	1,0595	0,4356	Equivalent
ESN + GA + SRG	B_J_Auto	-5,6845	-2,5000	0,6845	0,2368	Equivalent
ESN + GA + SRG	Ararma	-5,6845	-2,5000	0,6845	0,2368	Equivalent
ForecastPro	ForcX	-3,9345	-0,7500	2,4345	0,9929	Equivalent
ForecastPro	Holt	-3,5595	-0,3750	2,8095	0,9999	Equivalent
ForecastPro	THETA	-6,3095	-3,1250	0,0595	0,0585	Equivalent
ForecastPro	B_J_Auto	-6,6845	-3,5000	-0,3155	0,0204	ForecastPro is better
ForecastPro	Ararma	-6,6845	-3,5000	-0,3155	0,0204	ForecastPro is better
ForcX	Holt	-2,8095	0,3750	3,5595	0,9999	Equivalent
ForcX	THETA	-5,5595	-2,3750	0,8095	0,2960	Equivalent
ForcX	B_J_Auto	-5,9345	-2,7500	0,4345	0,1431	Equivalent
ForcX	Ararma	-5,9345	-2,7500	0,4345	0,1431	Equivalent
Holt	THETA	-5,9345	-2,7500	0,4345	0,1431	Equivalent
Holt	B_J_Auto	-6,3095	-3,1250	0,0595	0,0585	Equivalent
Holt	Ararma	-6,3095	-3,1250	0,0595	0,0585	Equivalent
THETA	B_J_Auto	-3,5595	-0,3750	2,8095	0,9999	Equivalent
THETA	Ararma	-3,5595	-0,3750	2,8095	0,9999	Equivalent
B_J_Auto	Ararma	-3,1845	0,0000	3,1845	1,0000	Equivalent

Table 34

Comparative results for the monthly M3 series.

Monthly Model	sMAPE
ForcX	14,88
SMARTFCS	15,19
Ararma	15,88
B_J_Auto	16,19
ForecastPro	16,44
Auto_ANN	16,88
ESN-GA-SRG	11,73

is possible to observe that the proposed model managed to obtain the first position in the comparison with the other six statistical models.

Table 25 shows the result obtained for the Post-Hoc test, where again all models are compared simultaneously, but this time, the significance level for all selected models is obtained. In this test it is possible to observe that only with half of the compared models it was possible to obtain statistically significant differences.

3.6.2. Time series results

Table 26 shows the average of the comparative results for the 12 series selected for the quarterly sampling group. Where the ESN-GA-SRG model again achieved the best result in terms of error.

Table 35

Comparative results for each monthly M3 series.

sMAPE Series	ForcX	SMARTFCS	Ararma	B_J Auto	Forecast Pro	Auto ANN	ESN-GA-SRG
1430	15,98	15,49	25,38	13,96	15,12	16,26	13,51
1460	50,54	51,39	63,25	63,25	56,99	48,82	38,52
1725	13,94	9,65	8,57	9,62	8,74	11,83	8,18
1900	2,51	4,20	3,39	2,93	2,80	3,04	2,49
1906	3,91	3,80	4,20	4,35	3,90	6,20	3,90
1912	6,40	3,96	6,86	9,20	6,39	10,24	5,14
1996	6,40	7,32	6,23	5,79	7,87	6,30	4,95
2001	18,87	19,56	18,02	17,19	17,06	21,33	17,02
2013	5,97	4,58	4,78	7,71	8,49	5,87	5,84
2016	2,68	3,30	4,12	2,07	7,85	4,43	1,55
2089	21,65	25,72	15,99	22,26	32,48	33,69	12,33
2579	29,77	33,30	29,83	36,02	34,75	32,20	27,38

Table 27 shows the sMAPE results for each of the 12 series, so the results in the 12 series can be compared individually.

Table 28 shows the configuration parameters of the ESN networks for the series that make up the M3 Quarterly set.

Next are the results obtained (tables 29 to 31) with the tests of statistical significance applied to the results of the M3 quarterly series. As in the previous cases, the value established for the level of significance, identified as p-value, was 5%, and the comparison of the different statistical models is performed using the accumulated sMAPE for each step of the forecast horizon.

Table 32 shows the result obtained for the Friedman test, where the six selected statistical models are compared simultaneously. It is observed that the proposed model managed to obtain the first position in the comparison with the other six statistical models.

Table 33 shows the result obtained for the Post-Hoc test, where again all models are compared simultaneously, but this time, the significance level for all selected models is obtained. In this test it is possible to observe that only with half of the compared models it was possible to obtain statistically significant differences.

3.6.3. Monthly series results

Table 34 shows the average of the comparative results for the 12 series selected for the monthly sampling group. Where the ESN-GA-SRG model again achieved the best result in terms of error.

Table 36

Parameters of the ESN-GA-SRG model for the M3 monthly series.

Series	Parameter					
	Jan	#U	%C	$\rho(W)$	sW^e	sW^{tr}
1430	4	227	60	0.1	0.1	0.8
1460	3	362	26	0.9	0.5	0.7
1725	12	462	20	0.4	0.8	0.1
1900	10	294	22	0.7	0.4	0.5
1906	10	468	34	0.6	0.6	0.1
1912	5	338	27	0.6	0.6	0.3
1996	12	299	42	0.3	0.9	0.2
2001	7	477	20	1	0.7	0.4
2013	8	451	70	0.8	0.8	0.3
2016	8	356	64	1	0.2	0.7
2089	7	187	44	0.4	0.8	0.2
2579	11	279	20	0.3	0.4	0.5

Table 37t-Student *t* test results for M3 series with M3 monthly sampling.

Model	<i>p</i> -value	Inferior Lim.	Superior Lim.	Conclusion
ForcX	1,92e-09	−3,0984	−2,1377	ESN + GA + SRG is better
SMARTFCS	4,57e-08	−3,1364	−1,9744	ESN + GA + SRG is better
Arama	8,38e-11	−3,3591	−2,4837	ESN + GA + SRG is better
B_J_Auto	3,57e-08	−4,3768	−2,7779	ESN + GA + SRG is better
ForecastPro	3,79e-06	−4,4304	−2,3061	ESN + GA + SRG is better
Auto_ANN	2,87e-03	−4,0375	−2,3756	ESN + GA + SRG is better

Table 38

Sync test results for M3 series with M3 monthly sampling.

Model	<i>p</i> -value	Inferior Lim.	Superior Lim.	Conclusion
ForcX	7,62e-06	−3,1499	−2,5713	ESN + GA + SRG is better
SMARTFCS	1,44e-04	−3,4051	−1,8988	ESN + GA + SRG is better
Arama	7,62e-06	−3,8944	−2,4571	ESN + GA + SRG is better
B_J_Auto	1,44e-04	−4,4714	−3,6137	ESN + GA + SRG is better
ForecastPro	0,0013	−4,5996	−3,2518	ESN + GA + SRG is better
Auto_ANN	0,0013	−4,2527	−3,2471	ESN + GA + SRG is better

Table 39

Wilcoxon test results for the M3 series with monthly M3 sampling.

Model	<i>p</i> -value	Inferior Lim.	Superior Lim.	Conclusion
ForcX	1,80e-04	−3,1230	−2,3463	ESN + GA + SRG is better
SMARTFCS	2,53e-04	−3,2027	−1,8308	ESN + GA + SRG is better
Arama	1,80e-04	−3,3559	−2,4642	ESN + GA + SRG is better
B_J_Auto	2,53e-04	−4,3875	−3,1440	ESN + GA + SRG is better
ForecastPro	4,19e-04	−4,4642	−2,9733	ESN + GA + SRG is better
Auto_ANN	3,00e-04	−4,0240	−2,7882	ESN + GA + SRG is better

Table 35 shows the results of sMAPE for each of the 12 series, allowing us to compare the results obtained in the 12 series individually.

Table 36 shows the configuration parameters of the ESN networks for the series that make up the monthly M3 set.

Below are the results obtained (tables 37 to 39) with the statistical significance tests applied to the M3 monthly series results. Just like in the previous cases, the value established for the level of significance, identified as *p*-value, was 5%, and the comparison of the different statistical models is performed using the accumulated sMAPE for each step of the forecast horizon.

Table 40 shows the result for the Friedman test, where the six selected statistical models are compared simultaneously. It is observed that the proposed model managed to obtain the first position in the comparison with the other six statistical models.

Experiments such as the competition series NN3 and M3 allowed us to perform a sensitivity analysis on different types of

series as well as experiments using the most used benchmarks in multi-step forecasting of time series.

The series' own characteristics and the proposed model achieved forecasts that may vary in terms of performance. In the case of the asymmetry coefficient, it was observed that time series with high values generate greater difficulty in forecasting. In the case of characteristics such as seasonality and stationarity, the proposed model managed to forecast considered with low errors and for the case of series with quarterly sampling, the results of the forecasts were equivalent.

Table 41 shows the result obtained for the Post-Hoc test, where again all models are compared simultaneously, but this time, the significance level for all selected models is obtained. It is observed that only with half of the compared models, it was possible to obtain statistically significant differences, for the other models ESN/GA/SRG has obtained a result that makes it significantly better.

Table 40

Results for the Friedman test for the M3 monthly series.

Model	ESN + GA+ SRG	ForcX	SMARTFCS	Arama	B_J Auto	Forecast Pro	Auto ANN
h = 1	4	5	6	7	2	1	3
h = 2	4	5	3	7	6	1	2
h = 3	1	6	7	4	5	3	2
h = 4	1	2	3	6	7	5	4
h = 5	1	4	2	3	7	5	6
h = 6	1	3	2	4	6	7	5
h = 7	1	3	2	4	6	7	5
h = 8	1	3	2	4	7	6	5
h = 9	1	4	3	2	6	7	5
h = 10	1	4	3	2	6	7	5
h = 11	1	3	4	2	7	6	5
h = 12	1	3	4	2	6	7	5
h = 13	1	3	4	2	6	7	5
h = 14	1	2	4	3	7	6	5
h = 15	1	2	3	4	6	7	5
h = 16	1	2	3	4	6	7	5
h = 17	1	2	3	4	5	7	6
h = 18	1	2	3	4	5	6	7
Sum	24	58	61	68	106	102	85
Average	1,3333	3,2222	3,3889	3,7778	5,8889	5,6667	4,7222
Position	1	2	3	4	7	6	5

Table 41

Results for the Post-Hoc test for the M3 monthly series.

Model	Model	Inferior Lim.	Average Dif.	Superior Lim.	p-Value	Conclusion
ESN + GA + SRG	ForcX	-4,0119	-1,8889	0,2341	0,1190	Equivalent
ESN + GA + SRG	SMARTFCS	-4,1786	-2,0556	0,0675	0,0651	Equivalent
ESN + GA + SRG	Arama	-4,5675	-2,4444	-0,3214	0,0122	ESN + GA + SRG is better
ESN + GA + SRG	B_J_Auto	-6,6786	-4,5556	-2,4325	0,0000	ESN + GA + SRG is better
ESN + GA + SRG	ForecastPro	-6,4564	-4,3333	-2,2103	0,0000	ESN + GA + SRG is better
ESN + GA + SRG	Auto_ANN	-5,5119	-3,3889	-1,2659	0,0001	ESN + GA + SRG is better
ForcX	SMARTFCS	-2,2897	-0,1667	1,9564	1,0000	Equivalent
ForcX	Arama	-2,6786	-0,5556	1,5675	0,9876	Equivalent
ForcX	B_J_Auto	-4,7897	-2,6667	-0,5436	0,0040	ForcX is better
ForcX	ForecastPro	-4,5675	-2,4444	-0,3214	0,0122	ForcX is better
ForcX	Auto_ANN	-3,6230	-1,5000	0,6230	0,3627	Equivalent
SMARTFCS	Arama	-2,5119	-0,3889	1,7341	0,9982	Equivalent
SMARTFCS	B_J_Auto	-4,6230	-2,5000	-0,3770	0,0093	SMARTFCS is better
SMARTFCS	ForecastPro	-4,4008	-2,2778	-0,1547	0,0262	SMARTFCS is better
SMARTFCS	Auto_ANN	-3,4564	-1,3333	0,7897	0,5128	Equivalent
Arama	B_J_Auto	-4,2341	-2,1111	0,0119	0,0524	Equivalent
Arama	ForecastPro	-4,0119	-1,8889	0,2341	0,1190	Equivalent
Arama	Auto_ANN	-3,0675	-0,9444	1,1786	0,8468	Equivalent
B_J_Auto	ForecastPro	-1,9008	0,2222	2,3453	0,9999	Equivalent
B_J_Auto	Auto_ANN	-0,9564	1,1667	3,2897	0,6692	Equivalent
ForecastPro	Auto_ANN	-1,1786	0,9444	3,0675	0,8468	Equivalent

4. Conclusion

In this article the ESN-GA-SRG model is presented, which considers the optimization of global parameters [68] for ESN networks in multi-step forecasting tasks of time series. The parameters were obtained using a Genetic Algorithm and then, the best one was selected. Reservoir topology used the SRG method that helped obtain interesting results in comparison to other forecast models.

The most critical parameters identified were the number of units in the reservoir, the reservoir connections, and the reservoir connections' weights. This identification is based on the behavior observed in the tests, where, for example, it was possible to identify networks with unchanged general parameters but with different responses (chaotic behavior). The initialization of weights can cause such a situation. On the other hand, it was also possible to identify networks with similar configurations of general parameters presenting similar responses (attractor behavior).

The main objective was to generate a greater compression of the functioning of the ESN networks, and thus to be able to take advantage of the potential offered by this type of models (Recurring Networks) to originate a model that has a joint optimization of global parameters and at the same time an interpretation of the behavior that contributes to the continuous research.

The degree of understanding of the echoes generated by the network during the pattern mapping process is quite low in the literature. Therefore, this article seeks to generate an approach that helps increase this understanding under the understanding of AutoML where two very important characteristics have been addressed: the optimization of hyperparameters and the selection of reservoir topologies.

In general, the results demonstrated that the ESN-GA-SRG model was able to improve the predictions made by other models and, it was possible to understand the behavior of the topologies used, such as the sensitivity in the configuration of the parameters,

where the results obtained confirmed that the simple changing of a parameter causes the results to change to unwanted forecast values.

Among the possible applications, it is considered that it can be applied to system identification problems in such a way that it allows characterizing and making various decisions depending on the final application.

As limitations, two special characteristics have been identified, the first, the computational cost in terms of hyperparameter optimization continues to be high, for which in future works it is proposed to address new theoretical forms for said task, secondly, the behavior of the reservoir through the analysis of SRG coefficient can be further exploited allowing for the characterization of the echoes which is still under study.

As future work, it is intended to expand the degree of integration by working on these models with quantum-inspired characteristics and thus generate new capabilities, also carry out much deeper explorations and with lower computational costs on the behavior of the reservoir and the way to be able to map it for different applications.

Data availability

No data was used for the research described in the article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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